ASSIGNMENT 1: EXPLORING THE PROBLEM SPACE

The goal of this assignment is to **research and define a research problem** by exploring the first stage of the Double Diamond design process.

- **Deliverables:** User-centric problem statement and argument/framing
- **Duration:** 2 hours

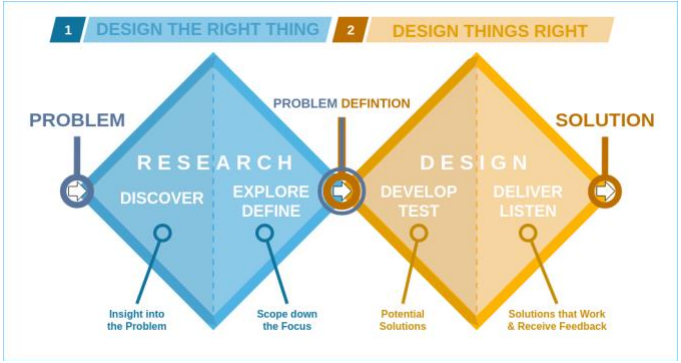
The design issues covered in this assignment are: (1) design process, (2) secondary research and (3) problem framing.

INTRODUCTION: WHAT IS DESIGN? HOW TO DESIGN?

Design is creative problem solving. Design is methods to research problems. It is ways to make sense of problems. Design is where ideas came from. Design is methods for generating ideas. It is methods for evaluating ideas. It is ways of communicating ideas. **Design is a process**.

In the first two assignments of this course, you will use the Double Diamond design process to **devise new (speculative) computing technologies and experiences** for a given context (i.e., physical setting and user). The Double Diamond design process is composed of two stages/diamonds – research and design. Each of the stages has a divergent (i.e., exploring an issue widely) and convergent (i.e., taking focused action) thinking phase, resulting in four phases:

- Discover: Understanding an issue by researching it.
- Define: Synthesize research and define the problem.
- Develop: Generating and prototyping multiple ideas to the previously defined problem.
- Deliver: Testing the solutions and reject/refine them in an iterative process.



ASSIGNMENT: DEFINING THE PROBLEM

We will use a 'Design Roulette' to randomly generate a design context for each team. We will generate three settings composed of pairs - physical space and user – and the team will be able to choose one of them (e.g., Hospital – Deaf children).

The overall goal is to design **new (speculative) computing technologies and experiences in a physical setting for a specific user group**. In this assignment you will **research possible challenges and opportunities for interactive technologies**.

Ideally, you would apply user research methods to better Discover people's problems, such as surveys, interviews, or observations. However, this is a **simulated assignment with a short timeframe**. Thus, **you will use secondary/desk research**, which does not communicate with people but leverages insights that others have gained from communicating with your user group, e.g., in books, articles, news, YouTube videos, etc. You can create a **mind map** to visually organize all possible problems that you collected.

After the Discover phase, you need to focus on a single problem and **define a user-centric problem statement**. The statement has three components: user, need, and goal. It follows a structure: **[users] need [need] to [goal]**. For example:

> *"Blind children need to be able to participate in collaborative classroom activities to increase their learning performance and social inclusion"*

Notice that **neither the need or goal refer to technology**. They focus on the current situation and problem space. Technology is part of the solution space (second Diamond – next class).

Once you have defined your problem statement, the final challenge is trying to explain the problem you are solving to other people. Thus, you will want to take all the knowledge you have and try to write a simple **argument that articulates the problem**. The argument should show the context, the scale of the problem and its consequences, i.e., it should show **why it is the problem important to solve**. For example:

> *"About one million blind and visually impaired (VI) children are educated in mainstream rather than special schools in the USA alone[1]. Current accommodations in mainstream classrooms include, for example, having a dedicated teaching assistant who sits with VI children through all classes and supports their learning activities[2]. Classroom content is usually made accessible through Braille, tactile diagrams or by using assistive technologies such as screen readers or screen magnification software. However, existing practices and technologies are designed to be used by children with visual impairments alone and not by their sighted peers, resulting in issues related to classroom participation, lack of collaborative learning, reduced social engagement, and potential for isolation."*

DELIVERABLES

At the end of the first studio, you should send one or more documents **via email** with:

1. **Problem statement**: user-centric problem statement following the structure: [users] need [need] to [goal].
2. **Argument**: a paragraph that articulates the problem you will try to solve.
3. **Other artifacts**: you can submit other documents that your team produced throughout the process, e.g., mind maps.

GRADING CRITERIA

Problem statement quality (50%): Has the team identified a meaningful problem? Did they follow a user-centric framing? Is the problem related to the physical setting and user group?

Argument quality (50%): Has the team shown the scale of the problem? Does the argument describe the consequences of the problem? Does it properly motivate the research problem? Does it answer why is the problem relevant?

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[1] https://www.disabilitystatistics.org/
[2] https://doi.org/10.1145/3173574.3173920